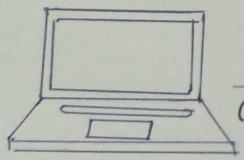
Web Browsing using OSI layers



User
(Web Browser)

Enters URL
(eg. www.google.com)

Encapsulation (Sender Side)

Application → DATA

Transport → SEGMENTS

Network → PACKETS

Data Link → FRAMES

Physical → BITS

7. Application layers
- HTTP /HTTPS /DNS
- Provide web services

6. Presentation layer
- encryption / decryption
- Data format / compression

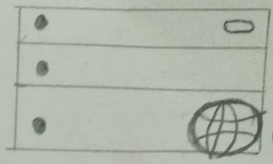
5. Session layer
- Establishes & manges session
- controls communication

4. Transport layer
- TCP / UDP
- segmentation & reassembly

3. Network layer
- IP addressing & Routing

2. Data link layer
- Ethernet / WiFi

1. Physical layer
- Cables / Fiber / Radio waves



Web Server
(Receives request)

Complete Web Browsing Flow

1. Enter URL
2. Application : HTTP/HTTPS + DNS
3. Presentation : Encryption / Formatting
4. Session : Session Management
5. Transport : TCP + Ports + Error Control
6. Network : IP + Routing
7. Data link : MAC + Frames
8. Physical : Bits Through Cable / Wi-Fi

↳ Internet

↳ Web Server